

CSC4182: Natural Language Processing

Semester II, 2024/25

Programming Assignment 01

Comparison of Tokenization and Embedding Methods on an NER Pipeline

In this assignment, you are required to build an NLP pipeline for the disease entity recognition task, where each token in a sentence must be assigned an appropriate named entity tag such as B-Disease (beginning of disease name), I-Disease (inside a disease name), or O (outside of a disease name) using the BIO tagging scheme.

Dataset:

The BC5CDR dataset is a labeled biomedical Named Entity Recognition (NER) dataset created using 1,500 PubMed articles. The dataset contains biomedical text annotated with entities such as diseases and chemicals. The dataset files required for this assignment are provided.

Example:

Token	Tag
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patients	O
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with	O
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diabetes	B-Disease
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mellitus	I-Disease
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were	O
------	---

treated	O
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NLP Pipeline:

Your pipeline should include tokenization, embedding generation, sequence modeling, and token classification.

In the embedding stage of the pipeline, you must compare the performance of FIVE embedding methods: **Word2Vec, GloVe, FastText, ELMo, BERT**

For tokenization, you may experiment with THREE tokenizers: **Whitespace tokenizer, NLTK tokenizer, Hugging Face BPE/WordPiece tokenizer**

For sequence feature extraction, you may use: **LSTM or BiLSTM**

For the classification layer, you may use: **Softmax or CRF**

You are encouraged to compare different pipeline combinations and analyze how tokenization and embeddings affect the final NER performance.

Deliverables:

1. An IPython notebook file with well documented steps.
2. A one-page report summarizing the F1-scores obtained for each pipeline, as shown below, and highlighting the best-performing pipeline.

Embedding	Tokenizer	F1-score
Word2Vec	Whitespace	
	BPE or WordPiece	
	NLTK	
GloVe	Whitespace	
	BPE or WordPiece	
	NLTK	
FastText	Whitespace	
	BPE or WordPiece	
	NLTK	
ELMo	Whitespace	
	BPE or WordPiece	
	NLTK	
BERT	Whitespace	
	BPE or Word Piece	
	NLTK	